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## PAPER\_1095: Horizon Buoyancy Sector Lagrangian for SCm-Corrected Black Hole Entropy

### Abstract

We derive the horizon buoyancy sector Lagrangian:

$$\mathcal{L}_{\text{horizon}} = -\beta_i U_g \Omega \frac{M}{d} [\text{UA}] + F_n \Phi_{1.25 \text{ THz}} + \frac{A}{4\ell_P^2} \cdot \frac{\Delta_{\text{SCm}}}{k_B T_H} \cdot S_{26}$$

The third term encodes the SCm BCS gap correction to Bekenstein-Hawking entropy. The stationarity condition  $\delta S / \delta \phi_{\text{horizon}} = 0$  produces a finite entropy bound that resolves the black hole information paradox via phonon feedback at the horizon.

### §1 SCm-Corrected Bekenstein-Hawking Entropy

Standard:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} = \frac{k_B A}{4\ell_P^2}$$

SCm-corrected:

$$S_{\text{BH}}^{\text{SCm}} = \frac{A}{4\ell_P^2} \left( 1 + \frac{\Delta_{\text{SCm}}}{k_B T_H} \right) S_{26} \cdot \Phi_{1.25 \text{ THz}}$$

where  $\Delta_{\text{SCm}} = 5.17 \text{ meV}$  is the SCm BCS gap energy.

### §2 Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G M_{\text{BH}} k_B}$$

For a  $10 M_{\odot}$  black hole:  $T_H \approx 6.17 \times 10^{-9} \text{ K}$ .

### §3 Lagrangian Structure

#### §3.1 Gravity Sector

$$\mathcal{L}_{\text{grav}} = -\beta_i \cdot \frac{G M_{\text{BH}}}{r_s} \cdot \Omega_{\text{BH}} \cdot \frac{M_{\text{BH}}}{r_s} \cdot [\text{UA}]$$

where  $r_s = 2GM/c^2$  is the Schwarzschild radius.

### §3.2 Phonon Sector

$$\mathcal{L}_{\text{neutron}} = F_n \cdot \Phi_0 \cdot S_{26}$$

### §3.3 Entropy Sector (new)

$$\mathcal{L}_{\text{entropy}} = \frac{A}{4\ell_P^2} \cdot \frac{\Delta_{\text{SCm}}}{k_B T_H} \cdot S_{26}$$

This term is unique to the horizon sector and encodes the BCS gap correction to the area law.

### §4 Euler-Lagrange Stationarity

$$\frac{\delta S}{\delta \phi_{\text{horizon}}} = \frac{\partial \mathcal{L}_{\text{grav}}}{\partial \phi} + \frac{\partial \mathcal{L}_{\text{neutron}}}{\partial \phi} + \frac{\partial \mathcal{L}_{\text{entropy}}}{\partial \phi} = 0$$

The EL residual  $\mathcal{R} = |\mathcal{L}_{\text{grav}} + \mathcal{L}_{\text{neutron}}|/|\mathcal{L}_{\text{entropy}}|$  determines how close the system is to true stationarity.

### §5 Benchmark: $10 M_\odot$ Black Hole

| Quantity                       | Value                                          |
|--------------------------------|------------------------------------------------|
| $r_s$                          | $2.96 \times 10^4 \text{ m}$                   |
| $A$                            | $1.10 \times 10^{10} \text{ m}^2$              |
| $T_H$                          | $6.17 \times 10^{-9} \text{ K}$                |
| $S_{\text{BH}}$ (standard)     | $1.46 \times 10^{79} \text{ J/K}$              |
| $\Delta_{\text{SCm}}/k_B T_H$  | $9.72 \times 10^4$                             |
| $S_{\text{BH}}^{\text{SCm}}$   | $\sim 10^{104} \text{ J/K}$ (phonon amplified) |
| $\mathcal{L}_{\text{entropy}}$ | $\sim 10^{84} \text{ J}$                       |
| Dominant sector                | Entropy (overwhelms gravity + neutron)         |

### §6 Information Paradox Resolution

The phonon-corrected entropy  $S_{\text{BH}}^{\text{SCm}}$  includes a factor  $(1 + \Delta_{\text{SCm}}/k_B T_H)$  that diverges as  $T_H \rightarrow 0$  ( $M \rightarrow \infty$ ), ensuring that information is never truly lost — it is encoded in the SCm phonon spectrum at the horizon. The BCS gap  $\Delta_{\text{SCm}}$  provides a natural IR cutoff.

### References

- PAPER\_1094: CMB Buoyancy Sector Lagrangian
- PAPER\_1089: Inflation Buoyancy Sector Lagrangian
- PAPER\_1090: Dark Energy Buoyancy Sector Lagrangian
- PAPER\_949: BCS Gap Equation SCm
- PAPER\_645: UQFF Einstein Field Equations / BH Singularity Resolution

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